Solving Proportions

You can use proportions to solve many kinds of problems. Use **cross products** to solve a proportion.

To solve using a proportion, be sure that the two ratios have corresponding units.

Example

Hide

At Bright's Movie Theater, 3 tickets cost $24. How much would 4 tickets cost?

Step 1

Write a ratio with two numbers from the problem. Include labels.

$\frac{3 \text{ tickets}}{\$24}$ = $\frac{1}{8}$ tickets/$

Step 2

Write a proportion, using a variable for the number you do not know.

$\frac{3 \text{ tickets}}{\$24} = \frac{4 \text{ tickets}}{\$p}$ → $(24)(p)\left(\frac{3 \text{ tickets}}{\$24}\right) = \left(\frac{4 \text{ tickets}}{\$p}\right)(24)(p)$ → $\frac{3p}{3} = \frac{96}{3}$

$p = 32$

Step 3

Multiply to get the cross products and find the unknown value.

$3 \times p = 24 \times 4$

$3 \times p = 96$

$p = 96 \div 3$

$p =$ **$32**

The cost of 4 tickets is **$32.**

Another way to set up a proportion for this problem is this: $\frac{\$24}{3 \text{ tickets}} = \frac{\$p}{4 \text{ tickets}}$ → $(4)(24) = (3)p$ → $p = \frac{4(24)}{3}$ = 32

It doesn't matter which number goes on top or bottom in the first ratio. What *does* matter is that both ratios are set up in the same order with corresponding units.

For example, in the problem above, writing the two ratios as

$\frac{\$24}{3 \text{ tickets}} = \frac{4 \text{ tickets}}{\$p}$ ✗

would *not* be a true proportion because they do not compare the same units in their numerators or the same units in their denominators.

Another way to express the same proportion is

$\frac{\$24}{\$p} = \frac{3 \text{ tickets}}{4 \text{ tickets}}$ ✓ or $\frac{p}{24} = \frac{4}{3}$ ✓

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Lesson 6: Ratios and Proportions

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